A4.1 Evolution and speciation

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-a4-1.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 151–155 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

SPECIATION

A4.1.8 Allopatric and sympatric speciation

pp. 151–152

In both, gene pools are isolated and then diverge. In **allopatric** speciation the populations are separated geographically; in **sympatric** speciation they share an area but are isolated by behaviour or timing.

A4.1.9 Adaptive radiation

pp. 152–153

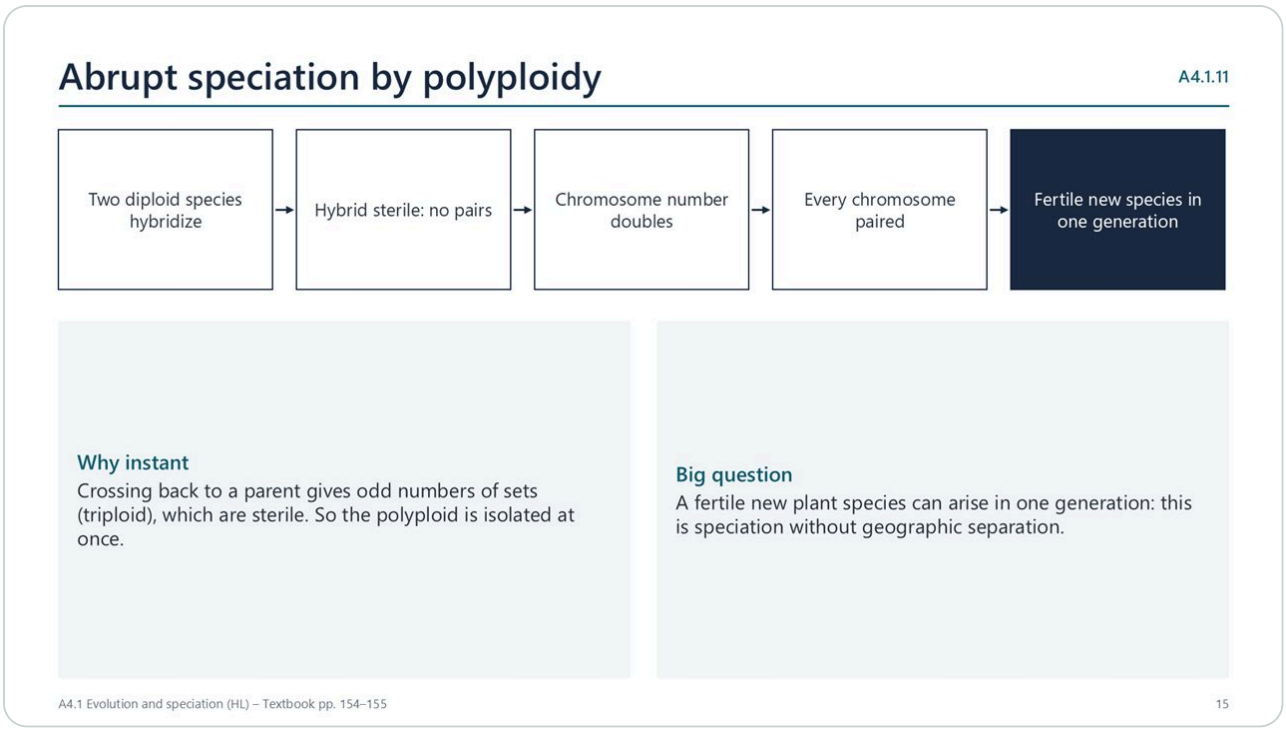
One ancestral species diversifies into many that fill different niches, as in Galápagos finches or bromeliads. Because they use different resources, close relatives can coexist without competing, which raises biodiversity.

A4.1.10 Barriers to hybridization

pp. 153–154

Species-specific courtship usually stops animals mating with other species. If hybrids do form, they are often sterile, as mules are, so alleles do not pass between species.

A4.1.11 Abrupt speciation in plants



Two species hybridize; the sterile hybrid doubles its chromosomes and becomes fertile. The new polyploid cannot breed successfully with its parents, so it is a new species at once. Many species of *Persicaria* (knotweeds and smartweeds) arose this way.

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